UNIVERSITÀ DEGLI STUDI DI PADOVA

AI-driven sustainable meal planning in institutional canteens

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Motivations

- €14B worth of food wasted in Italy every year[1].
- 30% of food produced in **school canteens** is wasted[2].

1. Waste Watcher International Observatory
2. 2019, Project Reduce, Italian Ministry for the Environment

Approach

- Follow users' **preferences** in terms of recipes and quantities;
- Prioritize **in-stock** food;
- Prioritize **low impact**, **seasonal**, **local food**;
- Promote food education through **variety**;
- Follow **WHO guidelines**.

Recipes selection

Select **recipes** in order to **satisfy** as many users as possible.

AI technique: **Ranked Pairs**

Foods selection

Select food in order to ensure:

- **Macronutrients** balance;
- Warehouse **availability**;
- **Low impact**, **seasonality** and **locality**;
- **Low cost**.

AI technique: **Genetic Algorithms**

Technical overview – Recipe blueprint

General template that defines the **structure** of a dish, not the exact ingredients.

Recipe – Cereals and legumes		
Ingredient	**Alternatives**	**Quantity**
Cereals	Rice **OR** Barley **OR** Wheat	50%
Legumes	Beans **OR** Chickpeas **OR** Lentils **OR** Peas	30%
Dairy	Parmigiano Reggiano **OR** Pecorino	15%
Oil	Olive oil	5%
Macronutrient		**Target**
Carbohydrates		60%
Proteins		20%
Fats		20%

Technical overview – Recipe blueprint

The **recipe blueprint** enables us to:

- **Separate** meal planning logic and specific food inventory data;
- Build a **fitness function** to evaluate a specific combination of foods with respect to a certain **recipe blueprint**, **warehouse availability**, and other **soft constraints**.

Technical overview – Fitness function

Given a **recipe blueprint**, the **fitness function** evaluates a certain combination of foods with respect to:

- **Macronutrients** distribution;
- `eco-score` and `seasonality-score`;
- Warehouse **availability**.

Technical overview – Gene modelling

A **gene** represents a **possible way of filling** a recipe blueprint.

Each gene corresponds to an **ingredient slot**, and encodes:

- The **specific food chosen** for that ingredient;
- A **quantity multiplication factor** to adjust proportions.

Recipe - Cereals and legumes - Genes			
(Rice, 1.1)	(Chickpeas, 0.9)	(Parmigiano Reggiano, 1.0)	(Olive oil, 1.25)
(Barley, 1.2)	(Lentils, 0.95)	(Pecorino, 0.9)	(Avocado oil, 1.1)
(Barley, 0.7)	(Chickpeas, 1.1)	(Parmigiano Reggiano, 1.2)	(Olive oil, 1.0)

Step 1 – Preferences collection

Preferences of the users					
User	Quantity	Choice 1	Choice 2	Choice 3	Choice 4
User 1	Small	Cereals and legumes	Salad and poultry	Pasta with vegetables	Salad and fish
User 2	Medium	Salad and poultry	Cereals and legumes	Salad and fish	Pasta with vegetables
User 3	Small	Salad and fish	Cereals and legumes	Salad and poultry	Pasta with vegetables
User 4	Large	Cereals and legumes	Pasta with vegetables	Salad and poultry	Salad and fish
User 5	Large	Salad and fish	Cereals and legumes	Pasta with vegetables	Salad and poultry
User 6	Medium	Salad and fish	Pasta with vegetables	Cereals and legumes	Salad and poultry
User 7	Small	Salad and poultry	Pasta with vegetables	Cereals and legumes	Salad and fish
User 8	Small	Cereals and legumes	Pasta with vegetables	Salad and poultry	Salad and fish
User 9	Large	Cereals and legumes	Salad and fish	Pasta with vegetables	Salad and poultry
User 10	Small	Pasta with vegetables	Cereals and legumes	Salad and fish	Salad and poultry

Step 2 – Recipes selection with Ranked Pairs

Selected recipes	
Recipe	**Quantity (kg)**
Cereals and legumes	81.00
Salad and poultry	52.00

Step 3 – Ingredients selection with Genetic Algorithm

Cereals and legumes					
Recipe	**Weight (kg)**	**Carbohydrates**	**Proteins**	**Fats**	**Price (€)**
Spelt, cooked	40.00	58.00%	12.00%	2.00%	45.00
Lentils, boiled	24.00	20.00%	9.00%	1.00%	30.00
Ricotta cheese, low fat	12.00	3.00%	6.00%	9.00%	22.00
Olive oil, extra virgin	4.00	0.00%	0.00%	100.00%	10.00
	80.00	38.88%	13.00%	6.75%	107.00
Brown rice, cooked	42.00	72.00%	7.00%	2.00%	38.00
Chickpeas, canned	25.00	27.00%	8.00%	2.50%	28.50
Parmigiano Reggiano	12.60	0.00%	30.00%	28.00%	49.00
Olive oil, extra virgin	4.20	0.00%	0.00%	100.00%	12.00
	83.80	45.58%	12.83%	10.06%	127.50

GitHub repository

Visit the GitHub repository to read and test the code.

`https://github.com/samuelemega/unipd-ai-canteens`

By combining **user preferences**, **nutritional science**, and **AI optimization**, we can make food services **smarter**, **healthier**, and **more sustainable** – one menu at a time.

Thank you!